TYPES OF ENERGY

There are many different kinds of energy, and most of them can be converted into other forms. For example, when you burn coal it changes the chemical energy stored in the coal into heat energy.

SOUND ENERGY
Energy we can hear, made when things vibrate.

KINETIC ENERGY
The energy objects have because they are moving.

ELECTRICAL ENERGY
The energy carried by electricity as it flows down a wire.

LIGHT ENERGY
Energy carried in electromagnetic waves.

CHEMICAL ENERGY
Released by a reaction between different chemicals.

NUCLEAR ENERGY
Generated by atoms splitting apart or joining together.

POTENTIAL ENERGY
Energy that is stored and yet to be released.

HEAT ENERGY
Energy stored or moved by molecules jiggling around.

HEAT

Heat is a form of energy, so when you heat something, you are increasing its stored energy. Objects store heat by jostling molecules or atoms inside them. Even large, cold objects can have heat energy.

HEAT IS USUALLY ON THE MOVE – IT TRAVELS ABOUT, SO COLD THINGS GET HOT AND HOT THINGS GET COLD

ICEBERGS
Icebergs are freezing cold but they still have some heat energy.

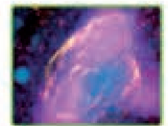
ELECTROMAGNETIC SPECTRUM

The Sun's heat and light is carried to Earth by electromagnetic waves. These are just part of a spectrum that includes radio waves, microwaves, and X-rays. All waves travel at the speed of light but they vary in wavelength, frequency, and energy.

MICROWAVES
Microwaves can be used to cook food.

INFRARED RAYS
Infrared radiation is a kind of "hot light". It shows up on thermal (heat-sensitive) cameras.

GAMMA RAYS
These are made when atoms split apart in nuclear explosions.



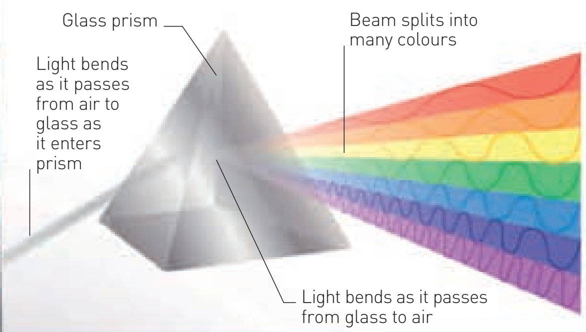
RADIO WAVES
Radio waves carry TV and radio signals between giant antennas such as this one.

ULTRAVIOLET
Sunlight contains ultraviolet waves. They tan your skin, but can also cause wrinkles and cancer.

X-RAYS
These have enough energy to pass through soft body tissue (like skin) but not bone.

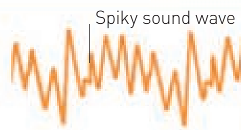
LIGHT AND COLOURS

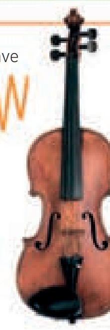
The light from the Sun looks white, but it is actually made up of lots of different colours. If you shine light through a prism, the whole spectrum of colours appears.



SOUND

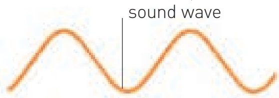
Sound is another form of energy that travels in waves. Louder sounds make bigger waves, while high-pitched sounds make waves that vibrate faster. The various noises we hear are produced by sound waves of different shapes and sizes.



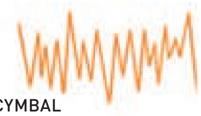


VIOLIN
When you play a violin, the strings vibrate, setting the air moving inside the hollow wooden case. A violin's sound wave is a sharp and spiky wave.

FLUTE
A flute produces sound when you blow into it, making waves inside the pipe. The sound waves are similar to a sine wave.



TUNING FORK
A tuning fork makes one simple, regular, up-and-down sound wave pattern called a "sine" wave. Each fork produces only one note.



A bigger cymbal vibrates a greater volume of air so it sounds louder

CYMBAL
Percussion instruments make sounds when you hit them. Their sound waves are more like a short burst of random noise (white noise) than a precise wave.

TINY SCIENCE

Our whole planet and all its people are made of atoms. The nucleus of an atom consists of protons and neutrons, and these are made of even smaller things called quarks. It is unclear what those are made of, but some scientists think that they may be vibrations of matter or energy, which scientists refer to as "strings". This science is known as quantum physics.

GREAT PHYSICISTS

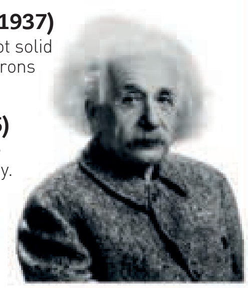
People have tried to explain our world and the Universe since ancient times. In the last 400 years, physicists have invented theories that underpin much of what we know.

ISAAC NEWTON (1643–1727)
Newton discovered that sunlight contains all the colours of the rainbow. He also devised the laws of gravity and motion.

ERNEST RUTHERFORD (1871–1937)
Rutherford proved that the atom was not solid but contained electrically charged electrons orbiting a nucleus.

ALBERT EINSTEIN (1879–1955)
Einstein discovered many things, but he is most famous for his theory of relativity.

RICHARD FEYNMAN (1918–88)
Feynman is best known for introducing the world to quantum physics.



ALBERT EINSTEIN